ECE510 Post-silicon Validation

TAP

Agenda

- Intro
  - IEEE Spec
  - TAP usage
  - Components
- Features
- Examples

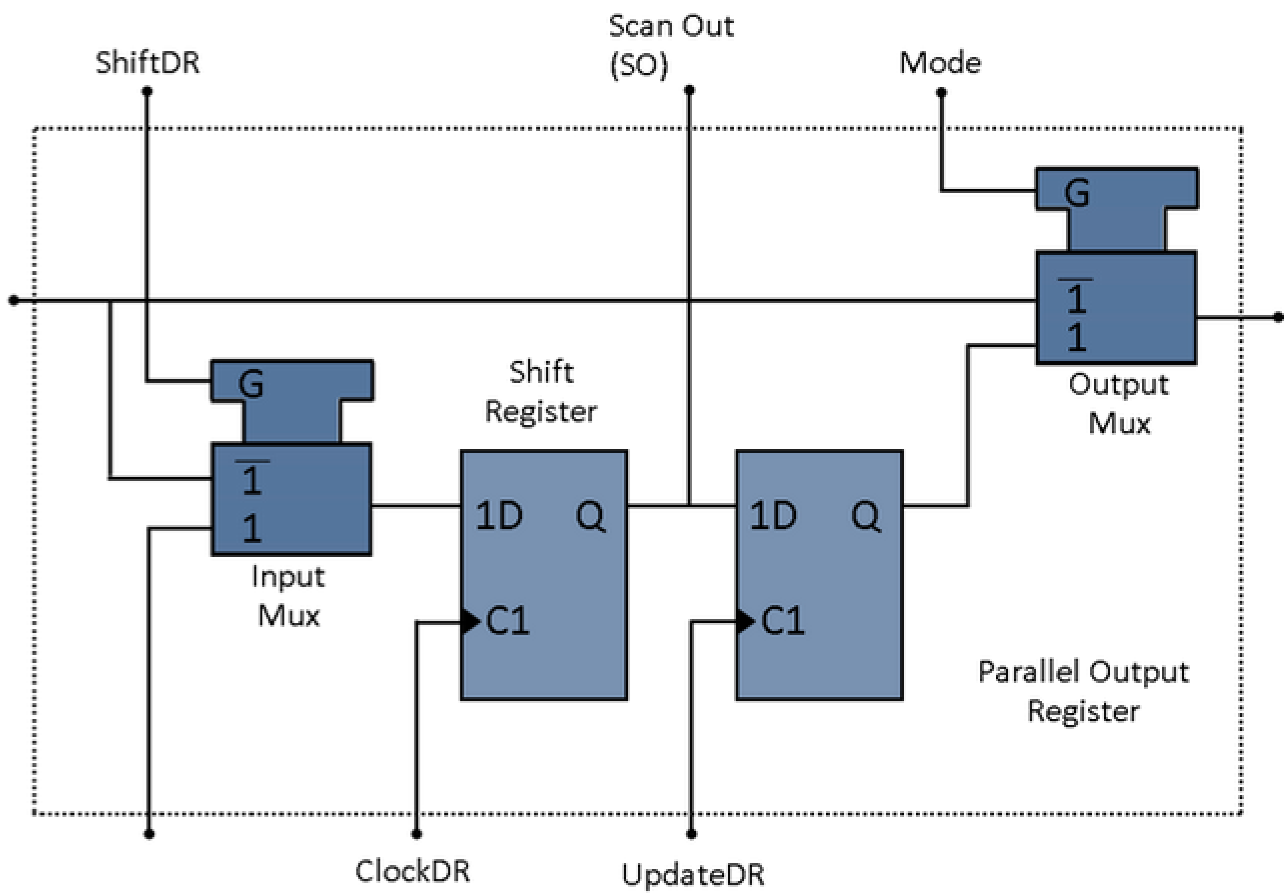
JTAG - Background

- Joint Test Action Group
  - Boundary Scan Architecture
  - Test Access Port

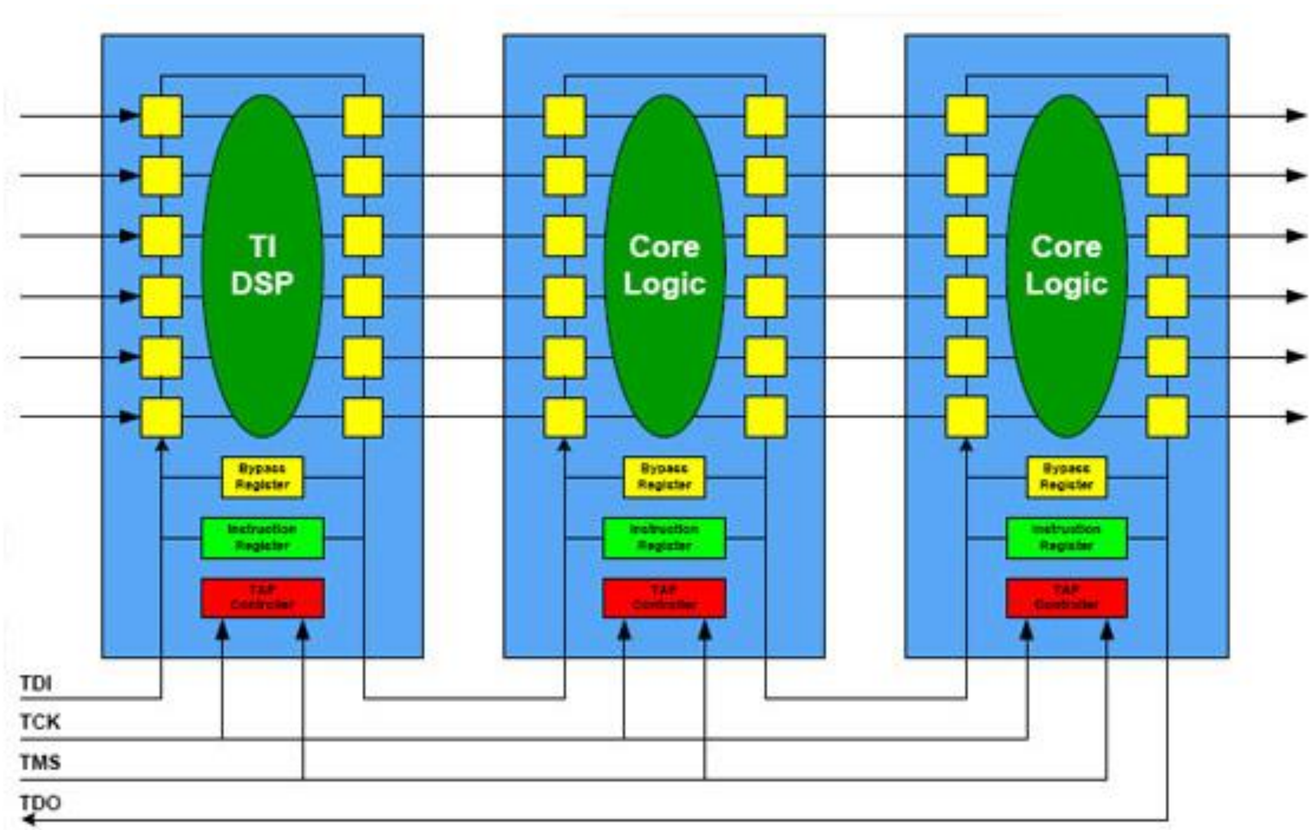
- History
  - IEEE1149.1 – 1990
    - Board level manufacturing test
  - IEEE1149.4 – 1999
    - Analog
  - IEEE1149.6 - 2003
    - Supports high speed data transmission, parallel data busses to serial buses with embedded clocks , can not longer be supported as boundary scan is a DC, single ended test
    - SERDES?
    - Uses differential signaling & AC coupling (fault tolerant)
  - IEEE1500
    - Device core access,

Bed-of-Nails tester

Boundary Scan Cells

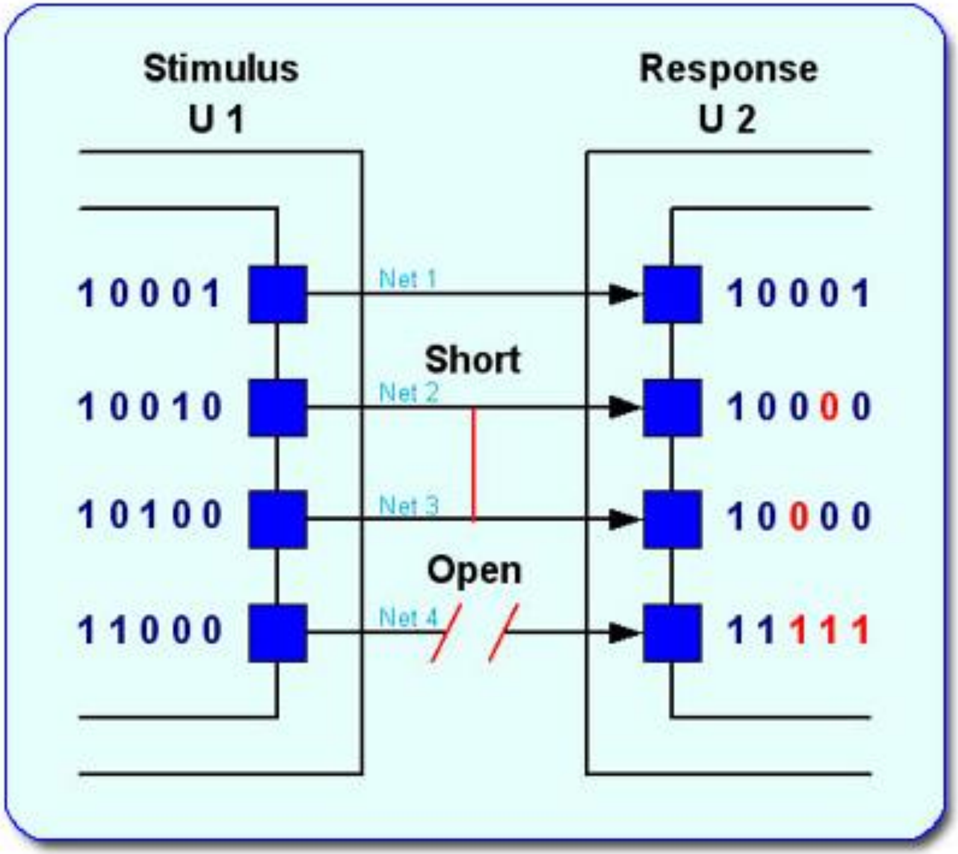


Board Level Boundary Scan



BSDL is used to describe the system

Boundary Scan example



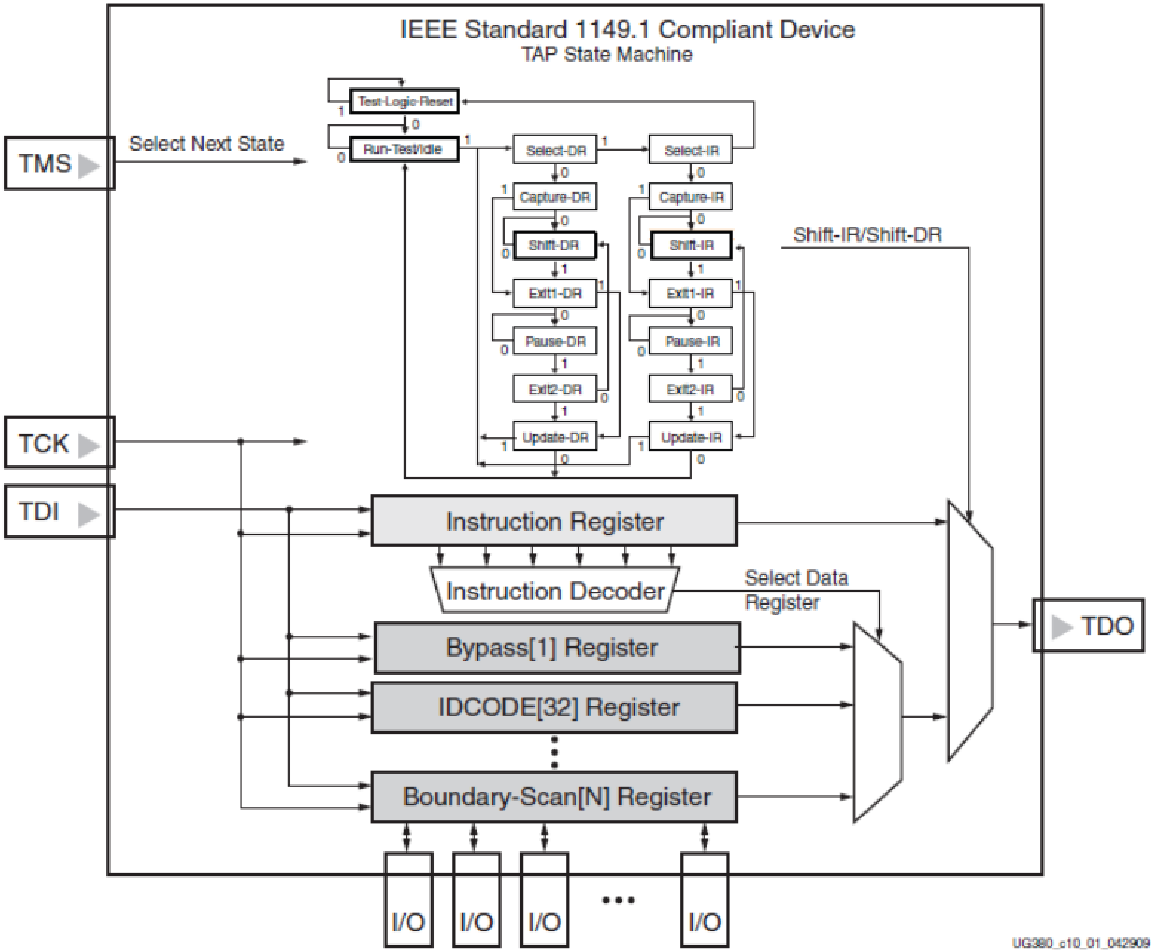
TAP

- Test Access Port
  - Also referred to as JTAG, the Joint Test Action Group
  - First defined in IEEE1149.1-1900 spec in 1985
    - Boundary scan description added in 1994
    - Other updates & minor changes along the way
- What's the need?
  - A standard serial interface for controlling the test logic
    - Obtain manufacturing info
    - Boundary scan, package interconnect
    - Internal scan paths
- Two types of registers:
  - IR: instruction register
  - DR: data register
    - Multiple data registers corresponding to different IR opcodes

TAP Usage

- SORT – tests are loaded through TAP or controlled via it
  - Tap bypass, idcode
  - EXTEST
  - Reset sequences, MBIST
  - Fuse overide
- CLASS
- Burn-in
- Speedpath – find speedpath, circuit marginalities through tests & debug hooks controlled through TAP
- Board test – test motherboard interconnect
- System debug
  - ITP – access individual registers

TAP


IEEE Standard 1149.1 Compliant Device
TAP State Machine
TMS
Select Next State
Test-Logic-Reset
Run-Test/Idle
Select-DR
Capture-DR
Shift-DR
Exit1-DR
Pause-DR
Exit2-DR
Update-DR
Select-IR
Capture-IR
Shift-IR
Exit1-IR
Pause-IR
Exit2-IR
Update-IR
Shift-IR/Shift-DR
TCK
TDI
Instruction Register
Instruction Decoder
Select Data Register
Bypass[1] Register
IDCODE[32] Register
Boundary-Scan[N] Register
TDO
I/O
I/O
I/O
I/O
UG380_c10_01_042909

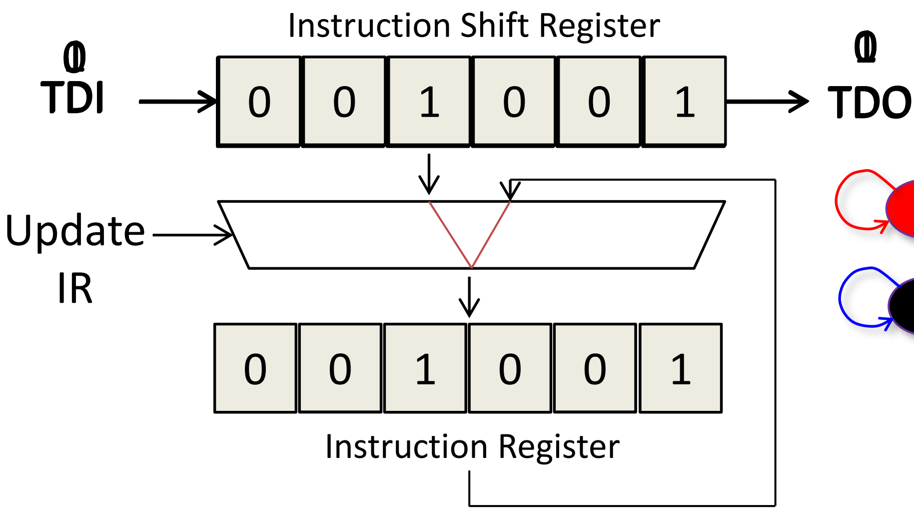
TAP Component

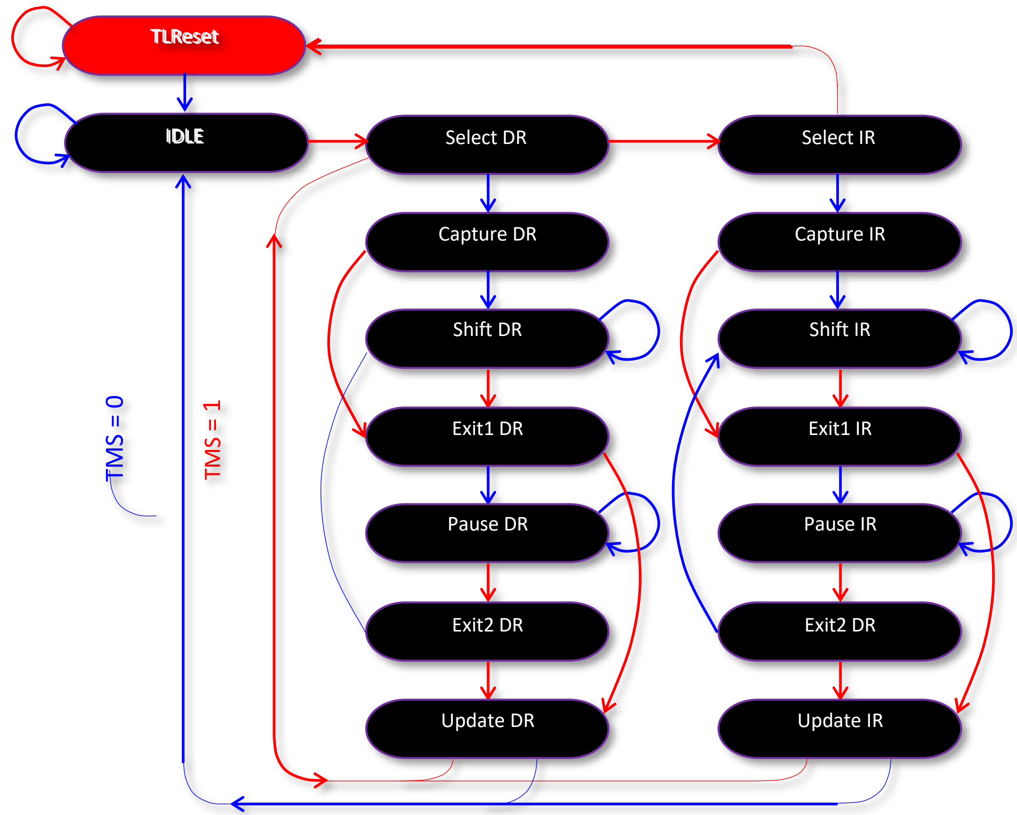
- TAP state machine
  - Controls the behavior of the JTAG system
- IR register
- Multiple DR registers
- Pins:
  - TCK – Test Clock
    - Frequency depend on the design
  - TDI – Test Data In
  - TMS – Test Mode Select – control the state
  - TDO – Test Data Output
  - TRST - optional

TAP – IR

- States we care about:
  - Capture IR
    - Preset data is latched into IR shift register
  - Shift IR
    - TDI data shifted into IR shift register at rising edge of TCK, TDI -> MSB of shift register
  - ExitIR
    - Transition out of shift IR, last TDI data will be latched
  - Update IR
    - Data in IR shift register are latched into IR register in parallel

TAP state transitions





TAP – DR

- States we care about:
  - Capture DR
    - Data from DR register are latched into DR shift register in parallel
  - Shift DR
    - Data is shifted out of DR shift register LSB first: LSB -> TDO
  - Update DR
    - Data in DR shift register are latched into DR register selected by the IR register (contains IR opcode) in parallel

TAP

- Common Instruction:
  - Bypass – insert one DD between TDI & TDO
  - Sample/Preload
    - Capture input/output pin values
    - Shift in input/output pin values
  - EXTEST
    - Test board level interconnect
  - INTEST
  - HIGHZ
    - Tri-state output pins into high impedance at the end of update IR, for leakage test, measure leakage to VDD for example
  - IDCODE
    - Read out component ID

EXTEST Instruction

EXTEST test sequence:
1, Shift DR (Chip 1)
2, Update DR (Chip1), Capture DR (Chip2)
3, Shift DR (Chip2)